ME4803/8803 EV Charging Homework

Please provide a source for each question and link it to a bibliography at the end. Ensure that every response includes a clear, thorough explanation. Include Name and Course Name/Section in title/file name. Failure to do so will result in deduction of points.

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Electric Vehicles and The Grid

1) Optimized Charging for your "City/State". All of the questions in this problem refer to the city/state that you have picked earlier in the semester.

a) What is the City/State that you have picked?
Shreveport, Louisiana

b) What is the cost of electricity (per kWh) in that city.
$0.12 per kWh avg. [1]

c) What are the Time of Use (TOU) rates? Usually these are done at the electric utility level, so you may have to choose a specific electric utility in the city (likely the largest utility). What are the details/times and rates for the TOU offering?
Rates for Southwestern Electric Power (SWEPCO) [2]
$0.1095; 5:00am-3:00pm
$0.1604; 3:00pm-7:00pm
$0.1095; 7:00pm-11:00pm
$0.0318; 11:00pm-5:00am

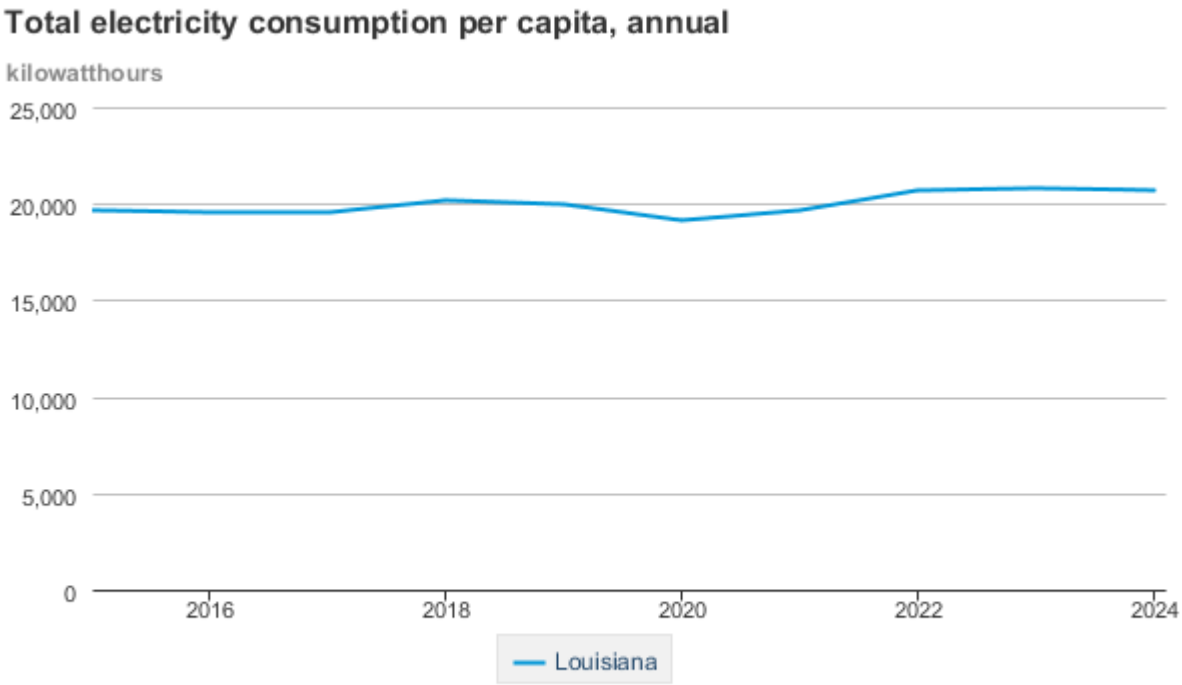
d) How does the utility generate its electricity (from what sources)? Can you locate the percentages and values for the region?

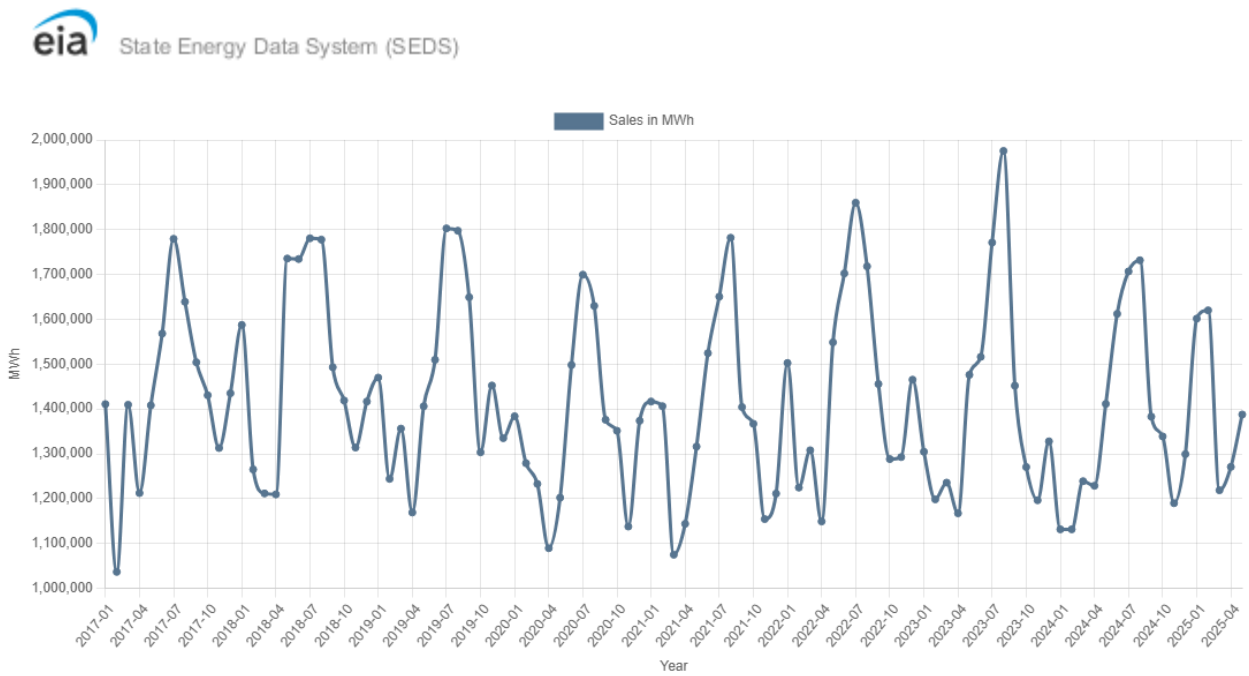
i) Coal	**53.52%**
ii) Nat Gas	**35.34%**
iii) Solar	**0%**
iv) Wind	**11.08%**
v) Hydro	**0%**
vi) Nuclear	**0%**
vii) Other	**0.07%**

[3]

e) Locate any information on how the carbon footprint for electricity generation changes by the hour in the region? If you can't locate specifics from the utility or ISO, what conclusions can you infer from question (d) above on how CO2 might vary by hour? Explain.
I was not able to find CO2 emissions directly for SWEPCO but I was able to use the EIA website to find total CO2 emissions for 2024. This value was 41,958 thousand metric tons. Using the following calculations we can determine the hourly rate of CO2 production for Louisiana, 41,958 thousand metric tons/year×1,000=41,958,000 metric tons/year and with 8,760 hours/year we can find that $\frac{41{,}958{,}000}{8{,}760} = 4{,}789$ **metric tons/hour of CO2. The 53% of coal power production that SWEPCO has indicates that baseline emissions are likely steady each hour as coal factories don't often ramp up or slow down, they remain mostly constant. [4]**

f) Locate any demand or generation graphs by hour for the region's electric grid. Explain / Interpret it.





[5]

I was not able to find by hour graphs or many graphs about Shreveport as a whole actually. The first graph shows the total electricity consumption per capita in the state of LA. The energy use is fairly

steady from 2015 to 2024 with about 20,000kWh per person per year. This means that the states demand is not trending in any direction, just mostly remaining constant. The second graph shows the sales in MWh over time on a monthly basis. This gives more of an understanding of what happens within the year. This graph shows much larger peaks during the summer months, mostly when people have their AC running full blast, especially in Louisiana. The years with the higher peaks and the lower dips indicate that the year was colder or hotter than usual. We can infer that in the peak months there is a low demand during the night because its cooler outside and a much higher demand during day hours when the sun is out and about.

g) Explain unique factors that you located relative to electricity generation for the utility? For example, is severe weather an issue? How much of the energy is used for cooling or heating homes? What is the percentage of residential electric demand vs industrial vs commercial?

SWEPCO's energy generation is heavily influenced by weather conditions with it being in LA. The IRP report that they released details that they use weather data, daily load, calendar effects, and temperature profiles to estimate the energy need at certain times throughout the day, month, and year. Severe weather is also a concern for the area, specifically winter storms, can affect operations and planning.

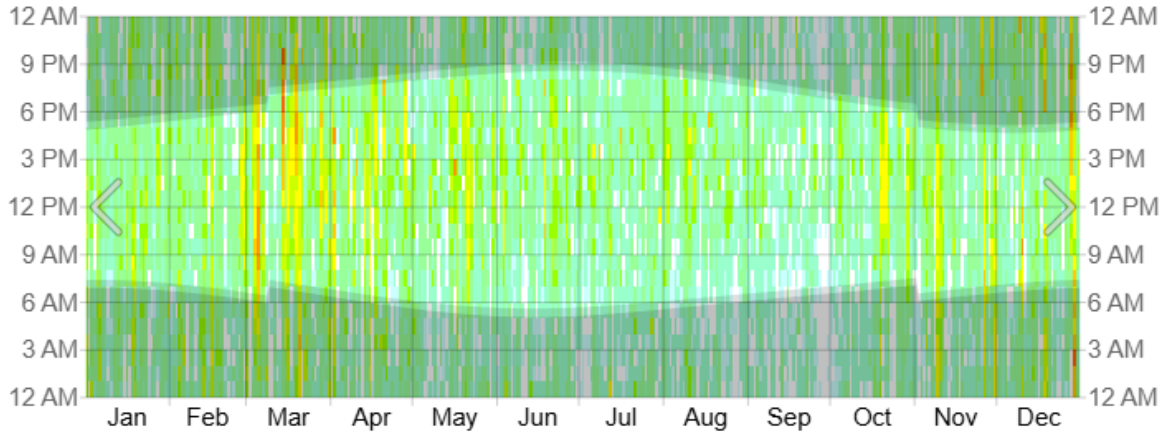
In terms of energy used by sector, SWEPCO indicates that the energy usage is weighted mostly to residential services being 3,029GWh, and commercial services being at 2,279GWh and industrial use being only 1,191GWh. This gives a percentage breakdown of 46.3% residential, 34.9% commercial and 18.2% industrial.

2) Using the information discovered above, what is an optimal charging algorithm for an EV located in that city/region? Feel free to get creative. For example, if your city has a high percentage of wind, you could write an algorithm that optimizes the charging based on wind speeds by the hour. You could optimize the algorithm for economics (i.e. price/cost), CO2, weather, electric load, etc. Write your algorithm approach below, and rationale. Provide calculations and in-depth explanation in your response.

I would want to write an algorithm to optimize the charging time so it's the most cost effective and optimize it for CO2 generation. I am going to use the TOU depicted above, and deeper research would need to be done for CO2 generation, hopefully finding a database about peak times would be most accurate however I will go off of the wind generation as SWEPCO has invested into that more than other renewable energies.

For the inputs to the system we would enter in the time buckets for the TOU:
$0.1095; 5:00am-3:00pm
$0.1604; 3:00pm-7:00pm
$0.1095; 7:00pm-11:00pm
$0.0318; 11:00pm-5:00am

We would also need to have wind generation approximation by hours:
According to this graph



[6]
It seems that the highest wind patterns are mostly in the middle of the day but with the wind picking up a lot in the mornings and evenings during the winter months and along with during march and April.
With this being said I think that during the summer months and march/April the highest wind times are 9am-3pm
And I think that the peak wind hours are between 12am and 6am and 6pm to 9pm for the winter months, specifically December

Lastly we would need to take into account when the vehicle would likely need to be used, the amount of energy it needs, and the charger limitations

This is the general formula for constructing this algorithm:

1. **Define the available charging window from plug-in time to departure time and determine the total energy needed.**
2. **Assign each hour a combined priority score based on:**

- **Cost priority: hours with lower TOU rate are preferred.**
- **CO_2 priority: hours with higher wind output are preferred (assumed lower CO_2).**
- **Layer A (cost filter): prioritize the cheapest TOU bucket first:**
    - **Best: 11:00pm–5:00am ($0.0318/kWh)**
    - **Next: 5:00am–3:00pm and 7:00pm–11:00pm ($0.1095/kWh)**
    - **Worst: 3:00pm–7:00pm ($0.1604/kWh)**
- **Layer B (CO_2 refinement inside each cost bucket): within each TOU bucket, prefer the hours with higher wind (cleaner electricity proxy).**
  For example:
    - **In winter (December): prioritize 12:00am–5:00am first (cheap + windy), then 6:00pm–9:00pm if more charging is needed.**
    - **In March/April and summer: within the $0.1095 bucket, prioritize 9:00am–3:00pm when wind is higher.**

3. **Schedule charging starting from the highest-priority hours until the required energy is met, while never exceeding the charger's maximum power per hour.**
4. **If no other times work based on departure and required charge time default to 3:00pm-7:00pm window as a last resort**

3) CO2 Calculation (and cost) for driving electric in your city/state.
   a. Using eia.gov, determine the cost per kWh and the CO2 per kWh for electricity.

.0880 $/kWh **0.927** lbs CO2/kWh

   b. Compare the two vehicles below in terms of cost and CO2 for 100 miles of driving. Use the average price of gasoline in the state. List your sources and values. Explain the differences and why you think they have the values listed.

      i. 2024 Toyota Corolla 2.0L Gasoline

$\frac{100\text{gal}}{35\text{mpg}} = 2.857\text{gal} * \$2.506 = \$7.16/100 \text{ Miles}$ $2.857\text{gal} * 19.592 \text{ lb/gal} = 55.98 \text{ lbs CO2/100 Miles}$
[7] [8] [9]

      ii. 2025 Hyundai Ioniq 6 EV (Standard Range)

$25\text{kWh} * \$0.088/\text{kWh} = \$2.20/100 \text{ Miles}$ $25\text{kWh} * 0.927 = 23.175\text{lbs CO2/100 Miles}$

[10]

The gas car cost is driven by gas price and mpg, and its CO_2 is high because every gallon burned releases 19.6 lb CO_2. The EV cost is usually lower in Louisiana because electricity is relatively cheap. The EV's CO_2 depends on the grid. Louisiana's grid emissions factor makes the EV cleaner than the Corolla, but the EV's emissions are not zero because the electricity still comes partly from fossil generation.

   c. Explain how the EV cost or CO2 compares to the national averages (how does driving in the state compare to the national average for greenhouse gas and cost)

National electricity
- **Avg retail price = 12.94 cents/kWh = $0.1294/kWh**
- **CO_2 factor = 785 lbs/MWh = 0.785 lbs/kWh**

National gas price (AAA national avg): $2.923/gal
EV (same car: 25 kWh/100 mi):
- **Louisiana cost: $25 \times 0.0880 = \$2.20$(cheaper)**
- **U.S. avg cost: $25 \times 0.1294 = \$3.235$(higher)**
- **Louisiana CO_2: $25 \times 0.927 = 23.18$**
- **U.S. avg CO_2: $25 \times 0.785 = 19.63$**

Gas Corolla (same mpg: 35):
- **Louisiana cost: $2.857 \times 2.506 = \$7.16$**
- **U.S. avg cost: $2.857 \times 2.923 = \$8.35$**

- **CO_2 per 100 miles is essentially the same everywhere for the same mpg (tailpipe CO_2/gal doesn't depend on state)**

4) Charge Couplers

4a) Identify each of the three couplers below:

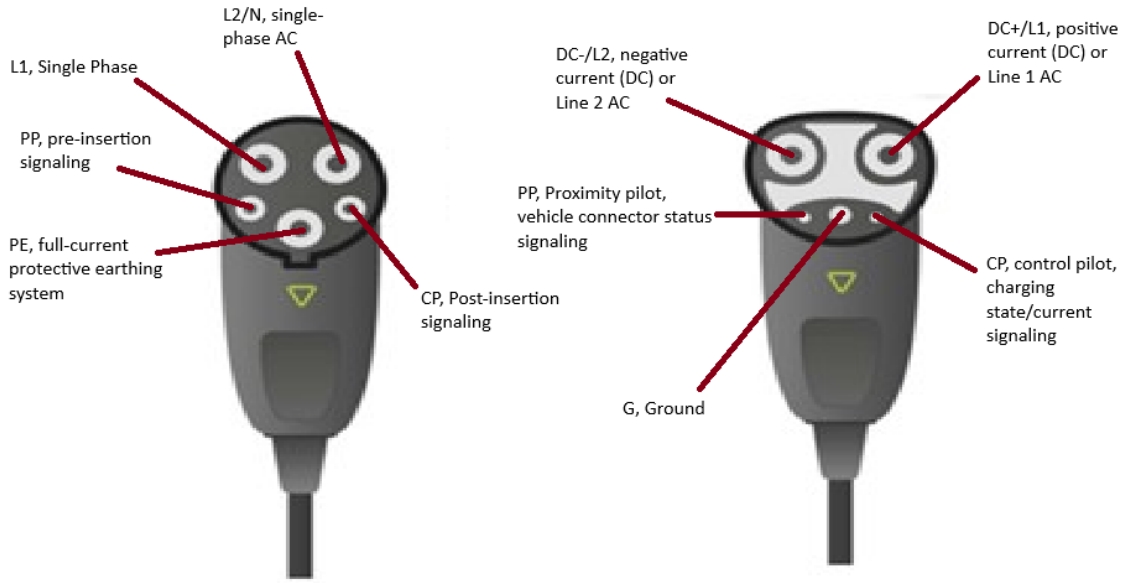
The far left is a Type 1 J1772

The middle is a Tesla Supercharger

The far right is a Type 2 Mennekes

Source: class lecture

4b) Label the pins for the Couplers Below:



[11] [12]

4c) Fill in the table for the voltage for each state below for J1772

State	Pilot Signal (Max Voltage)
EV Disconnected	+12 V
EV Connected	+9 V
EV Charge	+6 V

Source: 11

References

[1] https://poweroutage.us/solar/la/shreveport
[2] https://www.swepco.com/lib/docs/ratesandtariffs/Louisiana/LA_Tariff_Book_as_of_01-30-26.pdf
[3] https://findenergy.com/providers/southwestern-electric-power-company/#overview
[4] https://www.eia.gov/electricity/state/louisiana/
[5] https://findenergy.com/providers/southwestern-electric-power-company/
[6] https://weatherspark.com/h/y/11482/2025/Historical-Weather-during-2025-in-Louisiana-Missouri-United-States#Figures-WindSpeed
[7] https://gasprices.aaa.com/?state=LA&utm_source=chatgpt.com
[8] https://fueleconomy.gov/feg/bymodel/2024_Toyota_Corolla.shtml
[9] https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle?utm_source=chatgpt.com
[10] https://www.fueleconomy.gov/feg/bymodel/2025_Hyundai_Ioniq_6.shtml
[11] https://en.wikipedia.org/wiki/SAE_J1772
[12] https://en.wikipedia.org/wiki/North_American_Charging_Standard